AdifySkills

n8n Automation Mastery
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MENTOR COPY

🧩 Module 2 · Core Nodes & Data Mastery

DAY 05 of 25

Logic & Flow Control: IF, Switch, Merge, Loop

"Workflows jo khud decision lete hain."

⚡ AAJ HUM YEH BANAYENGE

Lead Scoring + Smart Routing

Webhook par lead aata hai, Code node score banata hai (budget, urgency, source), aur Switch hot lead ko Slack, warm ko nurture email, cold ko archive sheet mein bhejta hai.

IF | Switch | Merge | Loop Over Items | Wait

Naam: _____ Class date: _____

2 ghante ki live class · 7:00 PM – 9:00 PM IST

🧭 Aaj ka roadmap

Ek nazar mein poori class — kya, kyun aur kitne time mein.

💡 Yeh lesson kyun zaroori hai?

Real business processes seedhe nahi hote — "agar yeh, to woh". Flow control se n8n decisions leta hai.

🎯 Aaj ke baad aap yeh kar paoge

01 IF node & conditions (AND/OR)

🌍 Paise hain AUR dukaan khuli hai → shopping. (AND); chai YA coffee → kuch bhi chalega (OR).
⚙️ IF: {{ $json.paid }} is true AND amount > 0 → invoice branch.

02 Switch node — multi-route routing

🌍 Post office sorting: har letter pin code ke hisaab se alag bag mein.
⚙️ Switch on city: Delhi → rep A, Mumbai → rep B, fallback → manager.

03 Merge node modes: append, combine, by key

🌍 Do registers ko roll number se milana.
⚙️ Merge (matching fields: email) → CRM data + Sheet data ek item mein.

04 Loop Over Items (batching) & rate limits

🌍 Lift mein ek baar 8 log — baaki agli baar.
⚙️ Loop Over Items batch 50 + Wait 1s → API rate limit safe.

05 Wait node — delays & resume on webhook

🌍 Doctor: "3 din baad wapas aana".
⚙️ Wait 3 days → follow-up email; ya Wait "on webhook" → approval ke baad aage.

06 Filter, Remove Duplicates, Sort & Limit

🌍 Chhalni se kankad nikaalna, duplicate photos delete, marks ke hisaab se rank, top 10.
⚙️ Filter (city = Pune) → Remove Duplicates (email) → Sort (score desc) → Limit 10.

⏱ Class ka plan (2 ghante)

10m	35m	45m	20m	10m

`0:00-0:10` **Recap**

🪄 Cricket match se pehle warm-up karte ho na? Recap dimaag ka warm-up hai.

👩‍🏫 2 students se homework screen share karwao. Galti wala homework bhi dikhao — sabse zyada seekh wahin se aati hai.

`0:10-0:45` **Concept: Flow Control**

Flowchart → n8n mapping

🪄 Traffic police: green to chalo, red to ruko (IF). Chauraha: left, right, seedha (Switch).

👩‍🏫 Pehle real-life example se shuru karo, phir n8n screen dikhao. Har concept ke baad 1 student se apne shabdon mein dobara bulwao.

`0:45-1:30` **Live Build**

Smart lead router: hot/warm/cold → different actions

🪄 Aaj ki kahani yaad hai — 'Railway junction — kaun si train kis platform'? Ab wahi kahani n8n mein banayenge. Har scene ek node hai: ek-ek block jodo aur har block ke baad dekho sab chal raha hai ya nahi.

👩‍🏫 Har 10 minute par poocho "sabka green tick aaya?". Jaan-boojh ke ek galti live hone do aur use debug karke dikhao.

1:30-1:50 **Challenge**

Batch 500 records with rate limit

🪄 Practice match — jitna yahan girenge, utna asli match mein kam girenge.

👩‍🏫 Pehle 5 minute akele try karne do, phir hint do. Jo solve kare use screen share karwao.

1:50-2:00 **Q&A + Homework**

🪄 Ghar jaane se pehle bag check — sab samajh aaya? Homework diary mein likha?

👩‍🏫 Chat se top 3 doubts live solve karo. Homework ka pehla step class mein hi karwa do, taaki koi blank na baithe.

🧠 Concepts — aasaan bhasha mein

Har concept ke saath ek real-life analogy. Analogy yaad rahegi to concept bhi yaad rahega.

CONCEPT 01

IF

2 raaste: true / false.

🪄 **Analogy:** Traffic signal: green ya red

CONCEPT 02

Switch

3+ raaste, value ke hisaab se.

🪄 **Analogy:** Railway junction

CONCEPT 03

Merge

Alag branches ka data wapas jodna.

🪄 **Analogy:** Do nadiyan milkar ek

CONCEPT 04

Loop Over Items

Batch-by-batch processing — API rate limit ke liye.

🪄 **Analogy:** Ek baar mein 10 log lift mein

🔑 Aaj ka main idea

Flow control = workflow ka dimaag. IF 2 raaste banata hai (haan/naa), Switch kai raaste, Merge raaston ko wapas jodta hai, Loop bade data ko chhote batches mein chalata hai, Wait ruk ke baad mein chalata hai.

- IF: baarish ho rahi hai? → chhata lo, warna nahi
- Switch: marks 90+ A, 75+ B, baaki C
- Wait: chai ubalne tak 3 minute ruko

📖 Ek kahani se samjho: Railway junction — kaun si train kis platform

1 🚆 **Train aayi (lead)** = Webhook

2 🧮 **Ticket check (score)** = Code / Set

3 🔀 **Junction: 3 raaste** = Switch

4 🔥 **Express → Platform 1** = HOT → Slack sales

5 🐢 **Passenger → Platform 3** = COLD → archive

🪄 Ek line mein: “Traffic police: green to chalo, red to ruko (IF). Chauraha: left, right, seedha (Switch).”

🗺️ Aaj ka workflow: Lead Scoring + Smart Routing

Data left se right chalta hai. Har dabba ek n8n node hai.



💰 Isse kitna kama sakte ho?

Sales teams, B2B companies (lead scoring + routing) — aam taur par **₹10,000 – ₹25,000**. *(Andaaza hai; client aur scope ke hisaab se badalta hai.)*

🛠 Step-by-step build — n8n kholo aur saath chalo

1 ⚡ **Webhook**
Lead aata hai: budget, timeline, source

2 👨‍💻 **Code**
Score calculate: budget + urgency + source points

3 🔀 **Switch**
score ≥ 70 HOT, 40–69 WARM, baaki COLD

4 💼 **Slack**
HOT → #sales channel turant

5 📧 **Gmail**
WARM → nurture email sequence

6 📊 **Google Sheets**
COLD → archive sheet

✅ Har step ke baad **Execute step** dabao aur output check karo. Sab green ho to workflow **Active** karo.

🔑 Setup (import ke baad)

- ☐ Connect: **Slack (OAuth / bot token)**
- ☐ Connect: **Gmail (Google OAuth)**
- ☐ Connect: **Google Sheets (Google OAuth)**
- ☐ Slack mein #sales channel banao (ya Slack node ko Telegram se replace karo)

✅ Test kaise karein

1. Webhook node → "Listen for test event" dabao
2. Neeche wala curl command chalao (ya Postman / Hoppscotch se bhejo)
3. Budget badal-badal ke teeno routes test karo

⚖️ Pehle vs Ab — farak dekho

Yahi farak client ko dikhana hai. Time bachta hai = paisa bachta hai.

😩 Pehle (manual)

Har lead padh ke decide karna	5 min
Sahi insaan ko forward	3 min
Follow-up yaad rakhna	5 min
	13 min

⚡ Ab (n8n)

Score + routing auto	
Follow-up Wait node se	
	0 min

⏱ Har lead par 13 min bache, hot lead 1 minute mein sales tak.

⌨️ Aaj ka key snippet

Copy karke n8n mein paste karo — 📋 Select karke copy karo

```
const s = ($json.budget > 30000 ? 40 : 10) + ($json.timeline === "this week" ? 30 : 5);
return { ...$json, score: s };
```

📦 Sample data — input se output tak

📥 Input — 📋 Select karke copy karo

```
{
  "name": "Neha",
  "budget": 50000,
  "timeline": "this week",
  "source": "referral"
}
```

📤 Output — 📋 Select karke copy karo

```
{
  "name": "Neha",
  "score": 85,
  "label": "HOT",
  "routedTo": "slack#sales"
}
```

💬 Copy-paste prompts

ChatGPT / Claude mein seedha paste karo. PDF mein text select karke copy ho jata hai.

💬 Build it with ChatGPT / Claude — 📋 Select karke copy karo

```
Act as an n8n expert and teacher. Help me build this workflow in n8n step by step (n8n 1.x UI):

Goal: Webhook par lead aata hai, Code node score banata hai (budget, urgency, source), aur Switch hot lead ko Slack, warm ko nurture email, cold ko archive sheet mein bhejta hai.

Flow: Webhook (POST /score-lead) → Code (Score 0-100) → Switch (Hot / Warm / Cold) → Slack · Gmail · Sheets (Route karo)

For each node tell me: which node to add, exact settings, and the expressions to use. Then tell me how to test it and the 3 most common errors beginners hit.
```

💬 **Debug helper prompt** 📋 Select karke copy karo

```
I'm building the n8n workflow "Lead Scoring + Smart Routing" (nodes: Lead Webhook, Score Lead, Route by Score, Hot → Slack Sales, Warm → Nurture Email, Cold → Archive).
Node "<NODE NAME>" gives this error:
<PASTE ERROR>

Input JSON of that node:
<PASTE INPUT>

Explain the cause in simple words and give me the exact fix (settings or expression).
```

📥 Workflow ko n8n mein kaise laayein

1. **Workflow JSON** wale button par click karo (ya lesson page par 📋 **Copy workflow JSON** dabao).
2. Poora text copy karo (`Ctrl` + `A` , `Ctrl` + `C`).
3. n8n mein naya workflow kholo → canvas par click → `Ctrl` + `V` . Ya … menu → *Import from File*.
4. Upar wala Setup checklist poora karo aur **Test** dabao.

⚡ **Workflow JSON**
6 nodes · ready to import
`adifyskills.com/workflows/day-05-lead-scoring-router.json`

🌐 **Lesson page**
Copy buttons, quiz aur video links
`adifyskills.com/day/5`

🧯 Atak gaye? Yeh check karo

Error aana seekhne ka hissa hai — sab atakte hain. Neeche wale points se 90% problems solve ho jaati hain.

Agar yeh ho raha hai…	To yeh karo
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "Slack (OAuth / bot token)" connect karo.
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "Gmail (Google OAuth)" connect karo.
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "Google Sheets (Google OAuth)" connect karo.
Webhook/Form test par kuch nahi ho raha?	"Listen for test event" / Test URL use karo aur 2 minute ke andar request bhejo. Production URL sirf Active workflow par chalta hai.
"Sheet not found" / column khaali?	Sheet ID aur tab ka naam (capital/small bhi) check karo. Pehli row mein headers hone chahiye.

⚠️ Yeh galtiyan sabse common hain

- ! Merge node ka galat mode — data duplicate ho jaata hai
- ! IF mein string "10" vs number 10 compare karna
- ! Loop ke andar "done" output ko connect na karna

⭐ Pro tips — time bachane wali tricks

- Switch mein "Fallback output" on rakho — unexpected values ke liye
- Wait node se "3 din baad" follow-up — server restart par bhi resume hota hai
- Remove Duplicates (past executions) se same lead dobara process na ho

Phir bhi atke? Koi baat nahi 🙂
Screenshot + error message ke saath WhatsApp par poochho →

🏋️ Practice + Challenge

Padhne se zyada banane se seekhte hain. Ek-ek level upar jaao.

🛠 Live lab

Lead scoring + routing workflow: score ke hisaab se Sales Slack, nurture email ya archive.

📝 Homework

Do data sources ko Merge (by key) karke unified customer list banao.

- [ ] Homework ho gaya

BEGINNER
Budget 0 wale leads filter karo ☐

INTERMEDIATE
Round-robin: leads 3 sales reps mein baanto ☐

ADVANCED
Wait node + approval link se manager approval flow ☐

🧩 Aaj ka challenge — pehle khud try karo

🎯 Agar lead ka source "referral" hai to score ke bawajood hamesha hot maano.

💡 Hint 1
Yeh logic Code node mein ek extra line se ho sakta hai.

💡 Hint 2
return se pehle: agar source referral hai to score ko 70 ya usse upar kar do.

✅ Solution aakhri page par hai — pehle 5 minute khud socho 😉

🎯 2 minute ka self-check

Sahi option par gol ghera lagao. Answers aakhri page par — score sirf aapke liye hai.

Q1. 5 alag categories ke liye best node?

- (A) Switch
- (B) Merge
- (C) Wait
- (D) IF

Q2. API 1 sec mein sirf 10 requests leti hai — kya use karein?

- (A) Loop Over Items + Wait
- (B) IF
- (C) Merge
- (D) Sort

Q3. Do sources ko customer ID se jodna?

- (A) Filter
- (B) Merge (by key/matching fields)
- (C) Aggregate
- (D) Split Out

Q4. IF node ke kitne outputs?

- (A) 3
- (B) Unlimited
- (C) 1
- (D) 2

📖 Aaj ke naye shabd

Node
Workflow ka ek dabba/step. Har node ek kaam karta hai (email bhejna, data laana…).

API
Do apps ke baat karne ka tareeka. Ek app sawaal bhejta hai, doosra jawab (data) deta hai.

Switch
Kai raaston mein se ek chunna — jaise traffic police kisi gaadi ko alag lane mein bhejta hai.

Rate Limit
API ki speed limit — ek minute mein kitni requests allowed hain.

Webhook
Ek URL jis par koi doosra app data bhejta hai aur aapka workflow turant chal padta hai.

IF Node
Haan/naa wala decision — condition sahi to ek raasta, galat to doosra.

Merge
Do raaston ka data wapas ek jagah jodna.

👨‍🏫 Mentor notes — class kaise lein

Yeh page sirf aapki copy mein hai. Students ki PDF mein nahi jaata.

- [ ] Pehle paper par flowchart banwao, phir n8n mein convert
- [ ] Merge ke 3 modes ek hi data se live compare karo
- [ ] Batch challenge: 500 items ko 50-50 karke process

🗣️ Har block ke liye mentor tip

10m — Recap

2 students se homework screen share karwao. Galti wala homework bhi dikhao — sabse zyada seekh wahin se aati hai.

Class se poochho: Kal wale workflow mein sabse mushkil step kaunsa laga? · Homework mein kahan error aaya aur kaise theek kiya? · Kal ka ek node naam batao aur woh kya karta hai?

35m — Concept: Flow Control

Pehle real-life example se shuru karo, phir n8n screen dikhao. Har concept ke baad 1 student se apne shabdon mein dobara bulwao.

Class se poochho: 5 categories ke liye IF ya Switch? · 1000 emails ek saath bhejne mein kya problem? · Follow-up 3 din baad kaise?

45m — Live Build

Har 10 minute par poocho "sabka green tick aaya?". Jaan-boojh ke ek galti live hone do aur use debug karke dikhao.

Class se poochho: Is node ka output agle node mein kaise ja raha hai? · Agar yeh node fail ho to kya hoga? · Isko apne kaam/business mein kahan use kar sakte ho?

20m — Challenge

Pehle 5 minute akele try karne do, phir hint do. Jo solve kare use screen share karwao.

Class se poochho: 5 alag categories ke liye best node? · API 1 sec mein sirf 10 requests leti hai — kya use karein? · Do sources ko customer ID se jodna?

10m — Q&A + Homework

Chat se top 3 doubts live solve karo. Homework ka pehla step class mein hi karwa do, taaki koi blank na baithe.

Class se poochho: Aaj ka sabse useful node kaunsa laga? · Homework mein pehla step kya hoga? · Aaj ki class 1 line mein summarize karo.

✍️ Meri notes

Class mein jo naya laga, jo doubt aaya — yahan likho. Kal revise karna aasaan hoga.

⏭ Kal kya seekhenge — Day 6

Code Node: JavaScript for Automators

"Jahan node khatam, wahan Code node shuru."

☐ **Day 5 complete! 🎉**

Tick karo aur website par bhi ✅ dabao.

🔑 Answer key + useful links

Pehle khud try kiya? Ab check karo. Galat hua to explanation padho — wahi asli seekh hai.

Q1 → A
Switch

Q2 → A
Loop Over Items + Wait

Q3 → B
Merge (by key/matching fields)

Q4 → D
2

✅ Challenge ka solution

Score Lead code mein: if (b.source === 'referral') score = Math.max(score, 70);

📚 Official docs

- Flow logic ↗
- Merging data ↗
- Looping ↗
- Splitting with conditionals ↗

▶ Videos

- n8n if switch merge tutorial ↗
- n8n loop over items batching ↗

Doubt poochho
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